

Jhonathan Prieto Rojas, Ph.D.

E-mail: jprietor17@gmail.com | ORCID: [0000-0001-7848-1121](https://orcid.org/0000-0001-7848-1121) | ScopusID: [54983756500](https://scopusid.com/54983756500) |

Links: [Google Scholar](#) | [Website](#) | [KFUPM Website](#)

RESEARCH PROFILE SUMMARY

Electrical engineering researcher specializing in flexible and stretchable electronic systems for wearable sensing and energy harvesting applications. Focus on thermoelectric generators, mechanically compliant electronic architectures, and kirigami-inspired designs for flexible electronics. Author of peer-reviewed journal publications and U.S. patents in semiconductor devices and flexible electronics, with interdisciplinary experience spanning microfabrication, materials engineering, and electronic system integration.

Publications: 37 peer-reviewed journal papers

U.S. patents: 5

Total citations: ~1570 (Scopus), ~2080 (Google Scholar)

h-index: 20 (Scopus), 24 (Google Scholar)

EDUCATION

King Abdullah University of Science and Technology, Saudi Arabia

Jan. 2011 – Jun. 2014

Ph.D. Electrical Engineering, Electrophysics track (GPA: 3.84)

Dissertation Title: "Towards Flexible Self-Powered Micro-scale Integrated Systems"

Academic advisor: Prof. Muhammad M. Hussain, KAUST.

External. PhD committee member: Prof. John Rogers, UIUC, USA.

King Abdullah University of Science and Technology, Saudi Arabia

Sep. 2009 – Dec. 2010

M.Sc. Electrical Engineering, Solid State Devices track (GPA: 3.93)

Thesis Title: "Advanced Nanofabrication Process Development for Self-Powered SoC"

Academic advisor: Prof. Muhammad M. Hussain

Technical University of Munich, Munich, Germany

Oct. 2007 – Mar. 2008

Exchange Student, Electronics Engineering (Focus on VLSI design, hardware programming and Nanotechnology)

National University of Colombia, Colombia

Jan. 2003 – Mar. 2009

B.Sc. in Electronics Engineering (GPA: 4.3/5, Ranked Top 3 in Engineering College)

Thesis Title: "Study and Optimization of a Development Card for CPLD XC9572XL"

PROFESSIONAL & RESEARCH EXPERIENCE

Assistant Professor

Sep. 2015 – Aug. 2026

King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia ([QS #63 World; #1 Arab region](#))

- Conduct research in flexible and stretchable electronics, thermoelectric energy harvesting, and dust mitigation technologies.
- Teach undergraduate and graduate courses in electronics, digital design, and semiconductor devices and fabrication.
- Supervised 10 undergraduate senior design projects and 4 graduate research students.
- Serve on departmental committees related to ABET accreditation, teaching quality, faculty recruitment, and graduate admission.

Research Scholar

May 2021 - Aug. 2026

Interdisciplinary Research Center for Advanced Materials, KFUPM, Dhahran, Saudi Arabia

- Principal investigator on funded research projects in thermoelectric energy harvesting for wearable applications, and dust mitigation technologies.

Post-Doctoral Fellow

Oct. 2014 – Aug. 2015

King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Development of novel micro/nano-fabrication techniques for high-performance and cost-effective flexible electronics. (More-than-Moore applications)
- Development of Thermoelectric Generators (TEGs) by incorporation of nano-engineered materials and sustainable designs.

Visiting researcher**May 2015 – Jun. 2015**

University of Illinois at Urbana-Champaign, USA

Co-advisor: Prof. John Rogers

- Adaptation of transfer printing technology to improve the manufacturability of high-performance flexible electronics and enable 3D integration using flexible Si (100) fabrics.

Graduate Research Assistant**Sep. 2009 – Jun. 2014**

Integrated Nanotechnology Lab, KAUST, Thuwal, Saudi Arabia

- Development of novel micro/nano-fabrication techniques for high-performance and cost-effective flexible electronics. (More-than-Moore applications)
- Development of energy harvester/storage technologies by incorporation of nano-engineered materials and sustainable designs.

Visiting researcher, KAUST Global Collaborative Research (GCR) Program**Jun. 2010 – Aug. 2010**

The Pennsylvania State University, State College, USA

Co-advisor: Prof. Bruce Logan

- Design and successful fabrication of a micro-sized CNT-based Microbial Fuel Cell (MFC) with collaboration from the Materials Research Institute and Environmental Engineering Department.

Intern, "Product-to-Market" Department**Mar. 2008 – Jul. 2008**

Micronas GmbH, Munich, Germany

- Evaluated, diagnosed, and corrected functionality issues in the hardware and software of a MPEG-2 video converter board mounted on a Spartan3 FPGA.
- Delivered a fully functional board with technical documentation after 5 months of internship.

SELECTED JOURNAL PUBLICATIONS (Total: 37, 15 as first author, 5 as corresponding author, 7 cover pages)

1. **J. P. Rojas**, "Design and evaluation of LET-based soft hinges for stress mitigation in flexible copper interconnects", *Flex. Print. Electron.* 10, 035016 (2025). (Q2)
2. A. A. Alduais, A. B. Javaid, M. A. Syed, **J. P. Rojas**, "Analysis of Electrodynamical Screens with Fractal Designs for Enhanced Performance", *Arab. J. Sci. Eng.* 50, 17575-17585 (2025). (Q2)
3. **J. P. Rojas**, R. Almazyad, A. AlHayyah, A. Alruhaiman, M. Almusharraf, S. Al-Dharrab, H. Attia, "Self-Powered End-to-End Wireless Sensor Network for Geophysical Explorations", *IEEE Syst. J.* 19, 107-118 (2025). (Q1)
4. H. S. Daraghma, D. B. Ferry, S. G. Rao, M. A. Hawwa, M. A. Gondal, **J. P. Rojas**, "Materials & Design Approaches for Enhanced Performance of Mechanically Compliant Thermoelectric Generators (TEGs): A Review", *Smart Mater. Struct.* 33, 103003 (2024). (Q2)
5. M. S. Kim, A. S. Almuslem, W. Babatain, R. R. Bahabry, U. K. Das, N. El-Atab, M. Ghoneim, A. M. Hussain, A. T. Kutbee, J. Nassar, N. Qaiser, **J. P. Rojas**, S. F. Shaikh, G. A. Torres Sevilla, M. M. Hussain, "Beyond Flexible: Unveiling the Next Era of Flexible Electronic Systems", *Adv. Mater.* 36, 2406424 (2024). (Q1) **{Top viewed Article 2025 in Adv. Mater.}**
6. **J.P. Rojas**, M.U. Rehman, A. Hussein, A. E'mar, Y. AlRaei, H. AlZaher, M. Mohandes, "Kirigami-Enabled Wearable Health and Crowd Monitoring System", *Arab. J. Sci. Eng.* 47, 3583-3595 (2022). (Q2)
7. M.U. Rehman, W. Babatain, S.F. Shaikh, D. Conchouso, N. Qaiser, M. M. Hussain, **J. P. Rojas**, "Stress concentration analysis and fabrication of silicon (100) based ultra-stretchable structures with parylene coating", *Extreme Mech. Lett.* 41, 101052 (2020). (Q1)
8. M.U. Rehman, **J.P. Rojas**, "Optimization of compound serpentine-spiral structure for ultra-stretchable electronics", *Extreme Mech. Lett.* 15, 44-50 (2017). (Q1)
9. **J.P. Rojas**, D. Conchouso, A. Arevalo, D. Singh, I. G. Foulds, M. M. Hussain, "Paper-based origami flexible and foldable thermoelectric nanogenerator", *Nano Energy* 31, 296-301 (2017). (Q1)
10. **J.P. Rojas**, D. Singh, D. Conchouso, A. Arevalo, I. G. Foulds, M. M. Hussain, "Stretchable helical architecture inorganic-organic hetero thermoelectric generator", *Nano Energy* 30, 691-699 (2016). (Q1)
11. **J. P. Rojas**, G. A. Torres Sevilla, S. B. Inayat, A. M. Hussain, M. T. Ghoneim, S. Ahmed, M. M. Hussain, "Transformational Silicon Electronics", *ACS Nano* 8 (2), 1468-1474 (2014). (Q1)
12. J. E. Mink§, **J. P. Rojas§**, B. E. Logan, M. M. Hussain, "Vertically Grown Multiwalled Carbon Nanotube Anode and Nickel Silicide Integrated High Performance Microsized (1.25 µL) Microbial Fuel Cell", *Nano Letters* 12 (2), 791-795 (2012). §These authors contributed equally to this work. (Q1)

SELECTED CONFERENCES AND PROCEEDINGS (Total: 35, 12 as first author, 1 best presentation award)

1. **J. P. Rojas**, PV Dust Removal through Fractal-based Electrodynamics Screens. **KAUST Research Conference 2023: Sustainable Energy Materials and Technologies for a Low Carbon Future**. (Invited talk)(2023)
2. **J. P. Rojas**, A. Hussein, H. AlZaher and M. Mohandes, Stretchable Wearable Crowd Monitoring System, **2021 4th Inter. Symp. on Adv. Elect. and Comm. Tech. (ISAECT)** (Paper Presentation) (2021)
3. A. Alduais, A. Javaid, M. Syed, M. Alhrtomi, **J. P. Rojas**, Fractal Based EDS Design for Increased Solar Panel Dust Removal Efficiency, **2021 MRS Fall Meeting & Exhibit**. (Paper Presentation) (2021)
4. **J. P. Rojas**, Origami and Kirigami stretchable electronics. **IEEE NEMS 2020** (Invited talk) (2020)
5. **J. P. Rojas**, Organic-inorganic heterostructures for stretchable electronics. **Proceeding of SPIE, Micro-and Nanotechnology Sensors, Systems, and Applications XI** 10982, 1098215 (Invited – Session Chair) (2019)
6. **J. P. Rojas**, M. U. Rehman, M. A. Albettar, D. Conchouso, A. Arevalo, D. Singh, I. Foulds, M. M. Hussain. "Folding and stretching a thermoelectric generator"; **Proceeding of SPIE, Micro-and Nanotechnology Sensors, Systems, and Applications X** 10639, 106390E (Invited – Session Chair) (2018)

US PATENTS

1. Cylindrical-Shaped Nanotube Field Effect Transistor, M. M. Hussain, H. M. Fahad, C. E. Smith, **J. P. Rojas**. US Patent US9224813B2.
2. Apparatus, system, and method for on-chip thermoelectricity generation, M. M. Hussain, H. M. Fahad, **J. P. Rojas**. US Patent 9515245B2.
3. Integrated circuit manufacturing for low-profile and flexible devices, M. M. Hussain, **J. P. Rojas**. US Patent US9209083B2.
4. Method for Producing Mechanically Flexible Silicon Substrate, M. M. Hussain, **J. P. Rojas**. US Patent US9520293B2.
5. Flexible and foldable paper-substrate thermoelectric generator (TEG) **J. P. Rojas**, M. M. Hussain. US Patent 20170244019A1

RESEARCH PROJECT GRANTS

Projects as Principal Investigator (Total funding secured >320,000 SAR):

- | | |
|---|------------------------------|
| 1. <i>Interdisciplinary Research Center Grant (\$74,750 SAR)</i> | Feb. 2023 – Dec. 2024 |
| Development of Thermoelectric Generators on Flexible Substrates for Wearable Systems [Phase II] | |
| 2. <i>Interdisciplinary Research Center Grant (\$51,000 SAR)</i> | Jul. 2021 – Dec. 2022 |
| Development of Thermoelectric Generators on Flexible Substrates for Wearable Systems [Phase I] | |
| 3. <i>KFUPM Internal Funded Grant (\$146,190 SAR)</i> | Apr. 2017 – Jun. 2018 |
| Advanced Study of Microfabricated Stretchable Silicon-based Structures | |
| 4. <i>KFUPM Junior Faculty Research Grant. (\$48,180 SAR)</i> | Feb. 2016 – Oct. 2016 |
| Optimization of Spiral-based Structures for Stretchable Electronics | |

Projects as Co-Investigator:

- | | |
|--|-----------------------------|
| 1. <i>Georgia Tech-KFUPM Collaboration Grant</i> | May 2019- Nov. 2021 |
| Design and Implementation of a Wireless Geophone Network for Seismic Data Collection | |
| 2. <i>KACST Technology Innovation Center - EE2381</i> | Jan. 2018- Dec. 2020 |
| Solid-State Lighting: Optical-based Sensing for Advanced Health Monitoring System | |
| 3. <i>KAUST-KFUPM Collaboration Grant</i> | Oct. 2016- Oct. 2019 |
| Biometric data-based crowd management and risk mitigation for high population event | |

TEACHING AND ACADEMIC EXPERIENCE

Academic Programs Developed:

- Academic concentration (CX) in IC Design and Fabrications (EE, Phys, COE) (*Main Coordinator*)
- Academic concentration (CX) in Bioelectronics and Sensors (EE, Bio, COE) (*Team member*)
- Academic concentration (CX) in Mechatronics (ME, EE, COE) (*Team member*)

A CX is an interdisciplinary program consisting of 4 tailored courses offered to senior students to focus on a given specialized domain within their degrees.

Courses Developed & Taught:

- EE436 – Introduction to Micro & Nanoelectronics
- EE439 – Micro/Nanotechnology & Biosensors
- EE649 – Solid State Devices & Manufacturing (Graduate level)

Courses & Laboratories Taught:

- EE200 – Digital Logic Circuit Design (Multi-section Coordinator)
- EE201 – Electric Circuits I (Multi-section Coordinator)
- EE203 – Electronics I (Multi-section Coordinator)
- EE203L – Electronics I Lab (Multi-section Coordinator)
- EE204 – Fundamentals of Electrical Circuits (Non-EE Students)
- EE234 – Electronics & Microcontrollers (Non-EE Students) (Multi-section Coordinator)
- EE236 – Electronic Circuits (Non-EE Students)
- EE303 – Electronics II (Multi-section Coordinator)
- EE304L – Electronics II Lab (Multi-section Coordinator)

STUDENTS SUPERVISION

Undergraduate:

- Advisor for 10 undergraduate senior design projects related to dust mitigation technologies, electronic sensing systems for geological exploration and wearable applications.
- Advisor for over 12 undergraduate students in their Industrial Internships and Summer Training experiences.
- Academic advisor to approximately 20 undergraduate students per semester, guiding degree planning and academic performance.

Graduate:**M.Sc. Thesis Supervisor:**

- Title: Flow-Field-Based Routing Algorithms for Efficient PCB Design Automation. (In progress, Expected graduation: June 2026)
- Title: Optimization of Spiral-based Structures for Stretchable Electronics. (Completed, 2017)

M.Sc. Committee Member:

- Title: Dynamics of MDOF Metamaterial Resonators Attached to a Periodic Waveguide. (Completed, 2022)
- Title: Probabilistic Sampling Towards Tunable Autonomous Sensing. (Completed, 2022)

PhD Dissertation Supervisor:

- Title: Hybrid Dust Mitigation Techniques for Solar Panels in Arid and Semi-Arid Regions. (In progress, Proposal defended, Expected graduation: June 2027)
- Title: Development of Kirigami-inspired Thermoelectric Generators for Self-powered Sensing Applications. (In progress, Proposal defended, Expected graduation: Dec. 2026)

PROFESSIONAL SERVICE

Member, Assessment, Evaluation and Accreditation Committee**Jan. 2024 - Present**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Support the preparation of the Self-Study Report (SSR) for ABET accreditation.
- Coordinate the development and review of course specifications in the department.
- Assess the quality of teaching in the department.
- Support the development of guidelines for effective examination practices in the department.

Coordinator, Electronics Group, Electrical Engineering Dept.**Sep. 2021 – Aug. 2024**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Coordination of the development and revision of academic programs, new courses/laboratories, and the acquisition of equipment for academic purposes.
- Coordinate the evaluation process of faculty and graduate student applicants.

Member, Faculty Search Committee**Sep. 2023 – Aug. 2024**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Evaluation and interviewing of faculty applications, as well as lecturer and graduate assistant applicants.

Member, Academic Committee**Sep. 2023 – Aug. 2024**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Manage the modification, update, and restructuring of EE undergraduate and graduate programs.
- Supervise the introduction of new courses and programs.
- Assess the quality of all EE programs and propose enhancement plans.

Treasurer, IEEE Western Saudi Arabia EDS Chapter**Nov. 2019 – Aug. 2023**

- Maintaining financial accounts and overseeing all fundraising efforts of the chapter.

Member, Graduate Admission Committee**Sep. 2015 – Jun. 2023**

Electrical Engineering Dept., King Fahd University of Petroleum and Minerals, Dhahran, Saudi Arabia

- Evaluation of graduate applications and recommendations to the graduate program.
- Review and update graduate needs for the department and identify targets and directions.

Chairman, IEEE student branch**Jan. 2011 – Jun. 2014**

King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Coordination and organization of seminars, workshops, and social activities.
- Meeting and hosting academic visitors.
- Support of several technical and social activities in the electrical engineering department.

Vice-chairman, IEEE student branch**Sep. 2009 – Dec. 2010**

King Abdullah University of Science and Technology, Thuwal, Saudi Arabia

- Coordination and organization of seminars, workshops, and social activities.

EDITORIAL AND REVIEWING ACTIVITIES

Associate Editor**Jun. 2020 - Present**

Editorial duties in Frontiers in Electronics Journal (JIF 2.1)

Guest Editor**Oct. 2018 – Jan. 2019**

Journal of Sensors (JIF 2.1)

Co-edited a special issue in “Flexible and/or Stretchable Sensor Systems”.

Journal Reviewer for: Int. J. Mech. Sci. (JIF 9.4), Bioresour. Technol. (JIF 9), Compos. Struct. (JIF 7.1), Appl. Therm. Eng. (JIF 6.9), Adv. Mater. Technol. (JIF 6.2), Meas. (JIF 5.6), IEEE Electron Device Lett. (JIF 4.5), Rev. Adv. Mater. Sci. (JIF 3.9), Energy Technol. (JIF 3.6), J. Mech. Behav. Biomed. Mater. (JIF 3.5), IEEE Trans. Electron Devices (JIF 3.2), Flex. Print. Electron. (JIF 3.2), IEEE J. Flex. Electron. (CiteScore 5.2), Technologicas (Instituto Tecnológico Metropolitano, Colombia)

HONORS AND AWARDS

- Outstanding Alumni Award at International Level, Electrical and Electronics Engineers Association (AIEEUN), 60th Anniversary, National University of Colombia, 2022.
- *Second Place*: “Fractal-based Electrodynamical Screen”, 50K Challenge Award, KFUPM, May 2022. (Team advisor)
- Excellence in Teaching Award, College of Applied Engineering, KFUPM, Period 2020-2021.
- *First Prize*: “In-ear Health Monitoring System”, Hope Hackathon, Digital Health Track, Saudi Arabia, 2020. (Team advisor)
- *Awarded Second Prize* for “Novel Wafer Thinning Processing Based on XeF₂ Etching for Flexible Electronics”, on the KAUST 1st Graduate Research Symposium, PSE division, May 4th - 5th, 2011.
- *Best presentation Award* for “Carbon Nanotube based Micro-Sized Microbial Fuel Cell”, at the 5th Graduates’ Seminar at King Fahd University of Petroleum and Minerals, May 8, 2011.
- *KAUST Academic Excellence Award* (Fall 2010 - Spring 2011).
- *KAUST Graduate Fellowship and Provost Award* 2009. (In recognition to students with outstanding scholastic achievement and personal accomplishment).
- *German Academic Exchange Service (DAAD) - Young Engineers Scholarship* 2007. (Awarded for excellent academic performance and German language skills).
- Participation at *VIII Iberoamerican Olympiad of Chemistry*. Cuernavaca, Mexico. October 2-9, 2003.
- Honorary Mention, 2002 *Colombian Physics Olympiad*.
- Gold Medal (2002), Honorary Mention (2001), *Colombian Chemistry Olympiad*.

SKILLS AND MEMBERSHIPS

Languages	Spanish (native), English (fluent), German (Intermediate), Arabic (Basic).
Fab Skills	Photolithography, Wet Etch, Reactive Ion Etching, Atomic Layer Deposition (ALD), Plasma Enhanced Chemical Vapor Deposition (PECVD), Sputtering, Evaporation, Annealing, Scanning Electron Microscope (SEM).
IT Skills	COMSOL, Coventor, L-Edit, MATLAB, LATEX, C++, Visual Basic, VHDL, Labview, Flash, Assembler, MS Office, Windows, Linux, MAC.
Education tools	Blackboard, Blackboard Ultra, Poll Everywhere, Mentimeter.
Memberships	IEEE Electron Device Society Member, IEEE member.